HACK-A-STAT 2026

Group Name: Stat Attack

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Problem Statement Understanding:

The aim of this exercise is to check whether the study drug works better than the currently available standard of care by seeing how many patients survive without disease progression for 24 months.

Introduction

Lung cancer remains a major global health challenge, and evaluating the effectiveness of new treatments is essential to improving patient outcomes. This exercise analyses data from a **single-arm clinical trial** (meaning all enrolled patients received the study drug and there is no parallel control group within the trial) conducted to assess the efficacy of a study drug in lung cancer patients. The primary objective is to determine whether the study drug provides better outcomes than the existing standard of care, using **24-month event-free survival** as the key measure of effectiveness.

Background:

1. The trial included 200 patients diagnosed with lung cancer
2. An event is defined as either disease progression or death occurring during the follow-up period.
3. Censored Observations:
   i) Censoring occurs when the exact time of an event is not observed for a patient during the study period.
   ii) Censoring may occur:
   (1) a person does not experience the event before the study ends;
   (2) a person is lost to follow-up during the study period;
   (3) a person withdraws from the study because of some other reason

Exercise1:

Statistical approach used:

Part(a)

Data Source and Structure:

- Total number of observations: 200 patients.
- Each row represents one patient.
- There are 4 variables(columns) as:

Variable Name	Description
subject_id	Unique identifier assigned to each patient
recruitment_date	Date of patient enrollment and treatment initiation
event_or_withdrawal_date	Date on which a patient experienced an event or withdrew from the study
reason	Reason for study discontinuation, if applicable

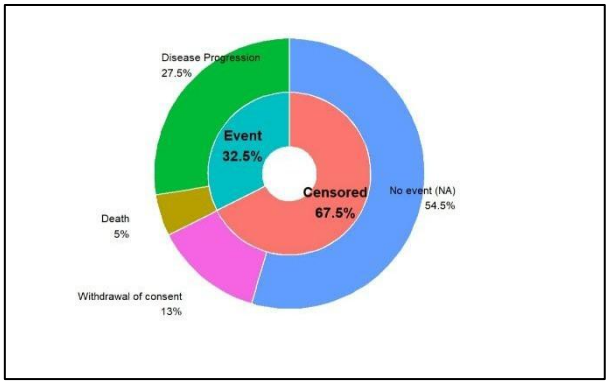
Missing Data Overview:

Missing values mean:

1) Missing values in event_or_withdrawal_date indicate that:
    a. The patient did not experience disease progression or death, and
    b. Did not formally withdraw from the study during follow-up.
2) Missing values in reason indicate:
    a. The patient either completed the study or was censored at the data cut-off date.

Dealing with the missing values:

1) We observed that cases for which the end date column was null were also the cases for which the reason column was also null and hence these cases were considered as censored cases.
2) Also, cases for which the reason was "withdrawal of consent" were dealt as censored.
   Hence, no sort of imputation or removal of cases/variables was required.



Feature engineering & derived variables:

To facilitate analysis of the primary endpoint, several new variables were derived from the raw dataset by converting date-based and categorical information into formats suitable for analysis.

- Follow-up duration (in months):
  A new variable was created by calculating the difference between the **event or withdrawal date** and the **recruitment date**, expressed in months. This variable represents the observed follow-up time for each patient. For patients without a recorded event or withdrawal date, the value was left as missing, indicating that the patient was censored at the data cut-off date.

One-Hot Encoding

Indicator for Outcome Within 24 Months:

Let d_{24} denote a binary **indicator random variable** defined as

$$d_{24} = \begin{cases} 1, & \text{if the difference in months is less than 24,} \\ 0, & \text{if the difference in months is greater than or equal to 24, or is missing (NA).} \end{cases}$$

Event Indicator (Failure Status):

Let δ denote a binary **indicator random variable** for event status, defined as

$$\delta = \begin{cases} 1, & \text{if the reason indicates disease progression or death (event),} \\ 0, & \text{if the reason corresponds to withdrawal of consent or is missing (censored).} \end{cases}$$

Month Event-Free Survival Outcome:

A final composite indicator was derived by combining the previous two binary variables. Patients who both experienced an event and did so within 24 months were coded as 1, indicating failure to achieve 24-month event-free survival. All other patients were coded as 0, representing either successful completion of 24 months without an event or censoring prior to event occurrence.

Let Y_{24} denote a binary **indicator random variable** for 24-month event-free survival status, defined as

$$Y_{24} = \begin{cases} 1, & \text{if an event occurs within 24 months,} \\ 0, & \text{otherwise.} \end{cases}$$

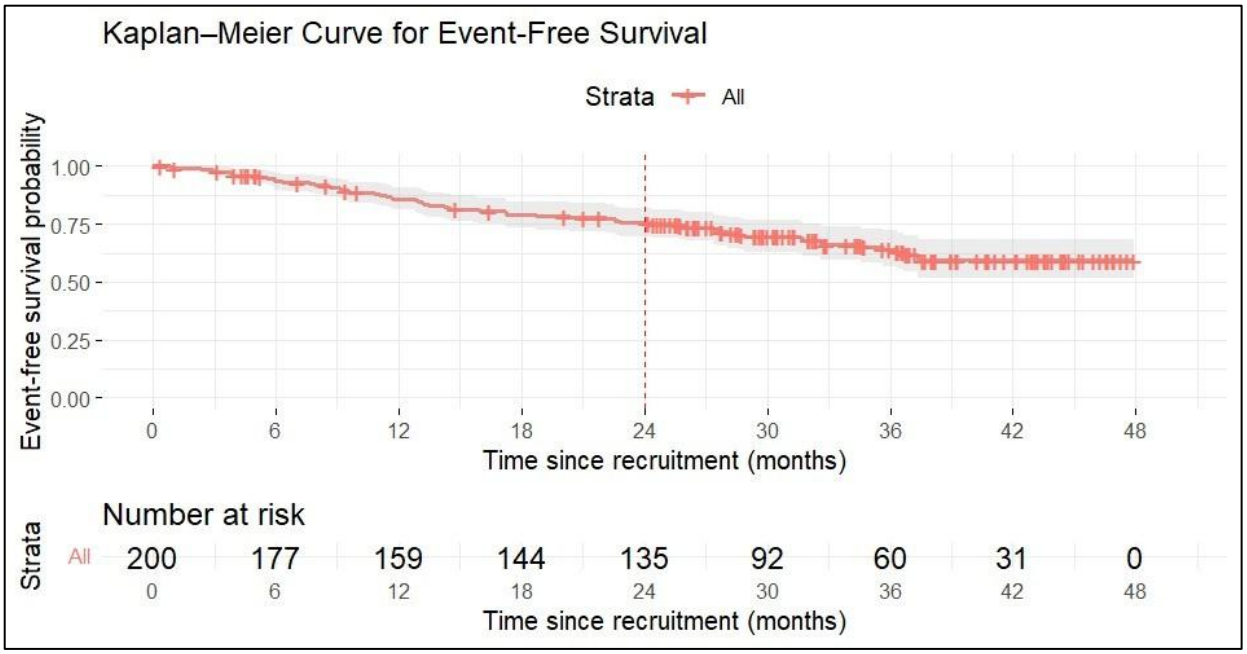
That is, $Y_{24} = 1$ when both $d_{24} = 1$ (month difference < 24) and $\delta = 1$ (event), and $Y_{24} = 0$ in all remaining cases.

These engineered variables directly align with the Kaplan–Meier framework, where survival probability is estimated as a function of follow-up time while properly accounting for censored observations.

Part(b)

To visually explore the primary endpoint, the Kaplan–Meier survival curve was used to estimate the event-free survival probability over time.

This plot provides a graphical representation of the proportion of patients remaining free from disease progression or death at each time point during follow-up.



- The Kaplan–Meier curve shows a **gradual and monotonic decline** in event-free survival over time, indicating that events occur progressively rather than in sudden clusters.
- At approximately **24 months**, the event-free survival probability is around **75%**, that is, greater than the standard of care event-free survival probability of 50%
- Increased censoring and a reduced number at risk after 24 months contribute to greater uncertainty in long-term survival estimates.

Part(c):

Survivor function:

The survivor function S(t) gives the probability that a person survives longer than some specified time t, that is, S(t) gives the probability that the random variable T exceeds the specified time t;

Where,

'T' is random variable for a person's survival time.

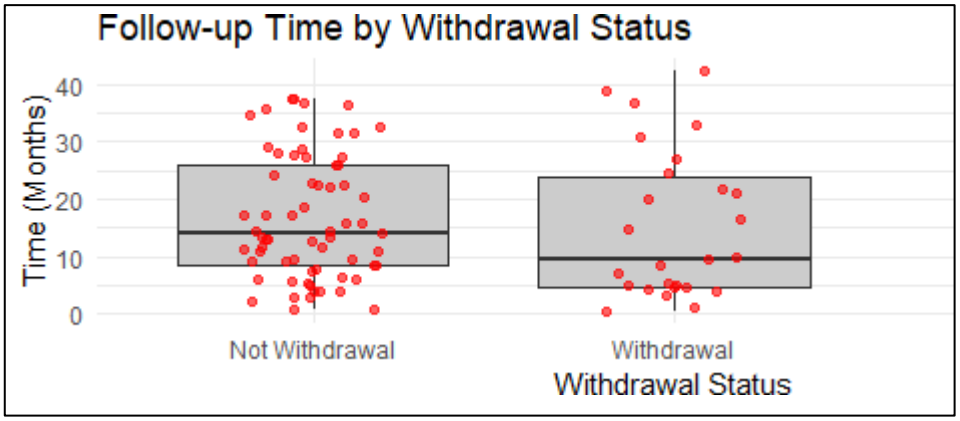
't' is specific value of interest for the random variable (here, t=24)

- at time t = 0, S(t) = S(0) = 1; that is, at the start of the study, since no one has gotten the event yet, the probability of surviving past time 0 is 1.

Censoring:

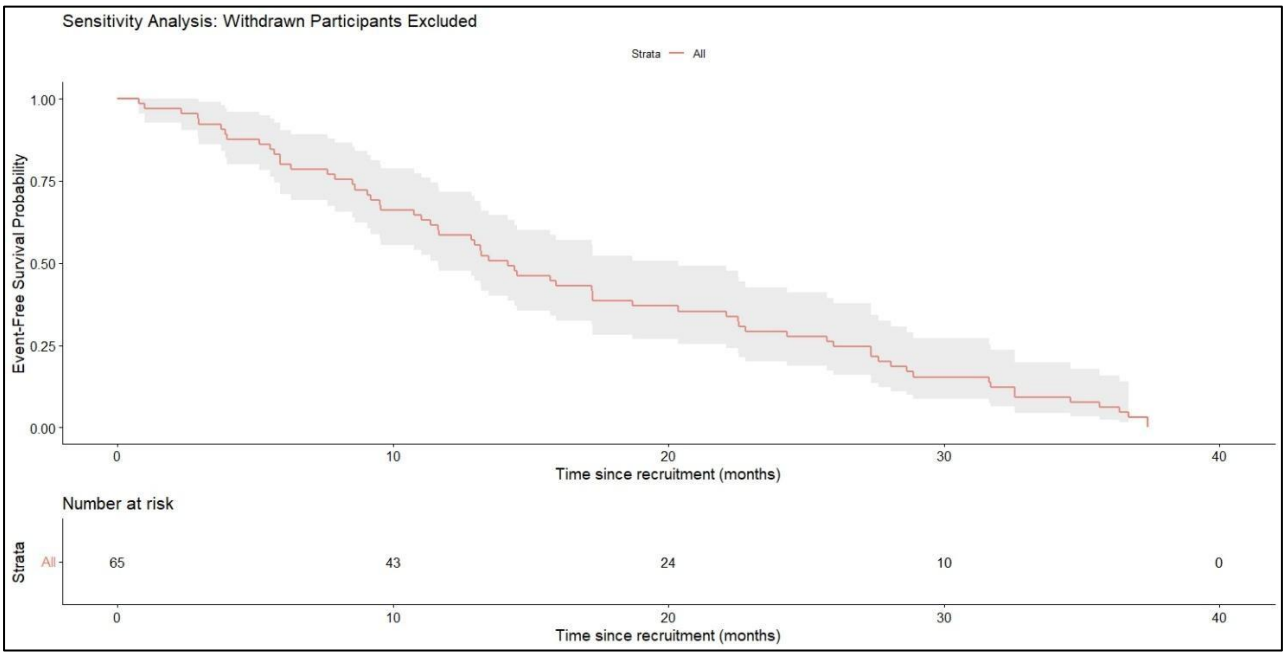
Our data is right censored.

Exploratory assessment of withdrawal cases v/s follow up time:



The comparison of follow-up time between withdrawn and non-withdrawn subjects indicates substantial overlap in the distributions of follow-up duration across both groups. Although the median follow-up time among withdrawn subjects is slightly lower, withdrawals are observed throughout the study period rather than being concentrated at early time points. This suggests that withdrawal is not strongly associated with time to event.

Further sensitivity analysis for informative censoring was done by plotting the KM curve for the complete censored data and withdrawal cases excluded.



The similarity between the Kaplan–Meier curves with and without withdrawn participants indicates that withdrawal-related censoring does not materially influence the estimated event-free survival. This suggests that the impact of withdrawals on the study conclusions is minimal, supporting the robustness of the survival estimates.

Hypothesis:

H_0: S(24) = 0.50, i.e, the 24-month event-free survival rate of the study drug is equal to 50%

v/s

H_1: S(24) > 0.50, i.e, the 24-month event-free survival rate of the study drug is greater than 50%

at α=0.05 level of significance

Assumptions:

• Independent Censoring:
Censoring is assumed to be independent of the event process. This means that patients who are censored have the same underlying risk of experiencing the event as those who continue to be followed.

• Independence of Survival Times
The survival times of different patients are assumed to be independent of each other, with no clustering or correlation between individuals.

• Date of recruitment is same as the date of the start of the drug administering**.**

• Consistent Survival Experience Over Time:

Patients enrolled earlier and later in the study are assumed to have the same survival experience, meaning that there are no major changes in treatment effectiveness or patient characteristics over the study period.

Kaplan-Meier estimator:

The Kaplan–Meier estimator is a non-parametric method used to estimate the survival probability over time in the presence of censored observations.

The Kaplan–Meier estimate of the survival function at time t is given by:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where,

- $\hat{S}(t)$ = estimated probability of surviving beyond time t
- t_i = ordered time points at which events occur
- d_i = number of events (e.g., disease progression or death) at time t_i
- n_i = number of individuals **at risk** just prior to time t_i
- The product is taken over all event times less than or equal to t

To quantify the uncertainty associated with this estimate, a confidence interval for the survival probability at 24 months was constructed using **Greenwood's formula** which provides an estimate of the variance of the Kaplan–Meier survival function by accounting for the number of events and the number of patients at risk at each observed event time.

Greenwood's formula:

$$\widehat{\mathrm{Var}}\left[\hat{S}(t)\right] = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}$$

Thus, a (1−α)% confidence interval for the survival probability at time t can be constructed as:

$$\hat{S}(t) \pm z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}\left[\hat{S}(t)\right]}$$

where, $z_{(1-\alpha/2)}$ is the standard normal quantile.

The decision:

- Reject H_0 if the lower bound of the confidence interval for S^(24) is greater than 0.50.
- Do not reject H_0 if the lower bound of the confidence interval is less than or equal to 0.50.

Calculations:

Event-free survival probability rate, S^(24) = 0.75

(1−α)% confidence interval for the survival probability = (0.69,0.81)

Conclusion:

Since lower-bound of the confidence interval(0.69)> 0.50 therefore , H_0 is rejected and therefore, we can conclude that 24-month event-free survival rate of the study drug is greater than standard-of-care 50%. Hence, 24-month survival rate of the study drug is significantly better than the standard of care currently available.

Exercise 2:

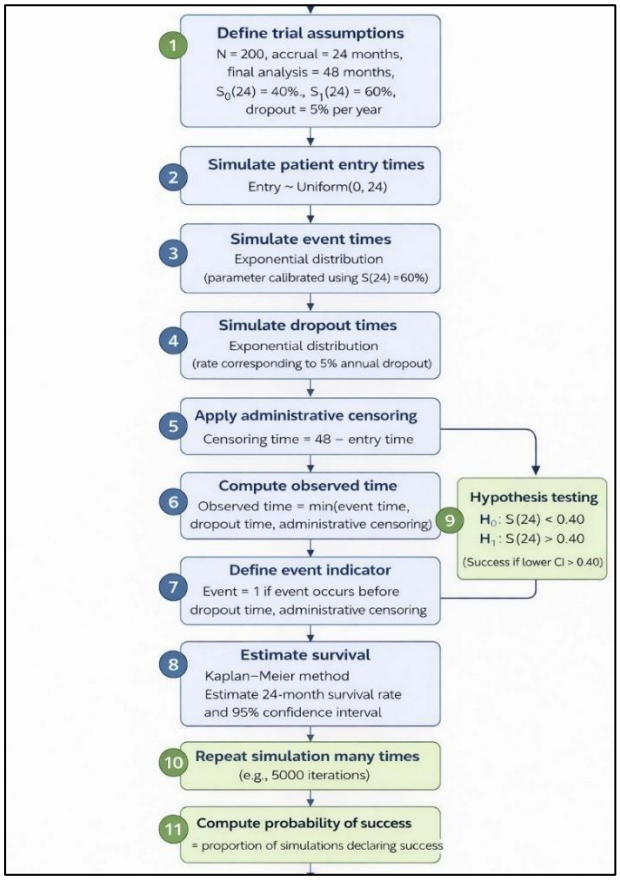
Problem Statement:

The pharmaceutical company plans to conduct another single arm trial using the same drug for patients diagnosed with liver cancer. The standard of care in this case is widely accepted to have a 24-month survival rate of 40%. The study team believes that the study drug under consideration will have a 24-month survival rate of 60% and plans to recruit 200 patients in the trial within 24 months. Note that since this is a single arm trial, patients will only be recruited in the study drug arm. Further, they anticipate an annual dropout rate of 5%. Under the study team's assumption, find the observed probability of success of the study drug being significantly better than the standard of care. You may have to conduct a simulation exercise to answer this question of interest.

Objectives:

1. To simulate the data of time at which the patient enters the trial.
2. To identify an appropriate distribution to simulate dropout time and event time of the patients
3. To identify which patients had an event, which ones dropped out, and which ones will be censored
4. To analyse the simulated data to see if the study drug is significantly better than the standard of care in terms of 24-month survival rate.

Methodology:



Study assumptions:

- The study is a single-arm clinical trial for liver cancer patients.
- Target sample size: 200 patients.
- Recruitment period: 24 months.
- Final analysis time: 48 months from trial start.
- Standard of care 24-month survival rate: 40%.
- Expected 24-month survival rate with study drug: 60%.
- Annual dropout rate: 5%.
- Significance level: $\alpha = 0.05$

Censoring Assumptions:

Survival probabilities were estimated using the Kaplan–Meier method assuming non-informative right censoring. Given the low dropout rate, any potential in formativeness of dropout-related censoring was considered negligible

Simulation of Patient entry time:

Patient recruitment times are simulated assuming uniform accrual.
Entry time follow a Uniform(0, 24) distribution (in months).
This reflects continuous and even patient enrolment over the recruitment period.
The 200 samples were generated for the 200 patients.

Simulation of Event (Survival) Time :

Event times are generated assuming an exponential survival distribution.The event rate (λ) is calculated to match the assumed 24-month survival rate of 60%:

$$\lambda = -\frac{\log(0.60)}{24}$$

Event times are simulated independently for each patient for the 200 patient

Simulation of the Dropout Time:

Dropout times are also modelled using an exponential distribution because of the constant rate.
The dropout hazard rate corresponds to a 5% annual dropout rate:

$$\lambda_{drop} = \frac{0.05}{12}$$

Dropout times are assumed to be independent of survival times.

Administrative Censoring:

Administrative censoring is applied at **48 months**.
For each patient, the maximum observable time is:

$$48 - \text{entry time}$$

Any event or dropout occurring after this time is censored.

Observed Time:

For each patient, the observed follow-up time is defined as the minimum of:

- Event time
- Dropout time
- Administrative censoring time

Event Indicator:

An event indicator is created:

1 if the event occurs before dropout and censoring.
0 if the observation is censored due to dropout or administrative censoring.

The R code output for the first 6 patients

	entry_time	event_time	dropout_time	admin_censor_time	observed_time	event
1	6.901860	84.716000	121.4928	41.09814	41.098140	0
2	18.919323	1.412276	13.5524	29.08068	1.412276	1
3	9.815446	61.239240	145.0743	38.18455	38.184554	0
4	21.192418	9.386792	296.1220	26.80758	9.386792	1
5	22.571215	82.288256	240.8488	25.42879	25.428785	0
6	1.093356	82.862014	486.1930	46.90664	46.906644	0

Estimation 24-Month Survival Probability:

The Kaplan–Meier estimator is used to estimate the survival function.
The estimated survival probability at **24 months**, denoted S^(24), is obtained from the Kaplan–Meier curve.

Hypothesis Testing Based on Confidence Interval:

Hypotheses

The objective is to evaluate whether the 24-month survival probability with the study drug is significantly higher than the standard of care.

H_0: S(24)=0.40

H_1: S(24)>0.40

Estimation

The 24-month survival probability, S^(24), is estimated using the Kaplan–Meier estimator**.**

The standard error of S^(24) is obtained using **Greenwood's formula**.

```
> cat("Estimated S(24):", round(S24_hat, 3), "\n")
Estimated S(24): 0.676
```

Construction of Confidence Interval

A one-sided 95% confidence interval for S(24) is constructed using a log–log transformation of the Kaplan–Meier estimator to ensure bounds lie within [0,1].

The lower confidence limit is given by:

$$\mathrm{LCL}_{95\%} = \exp\left(-\exp\left[\log\{-\log(\hat{S}(24))\} + z_{0.95} \cdot \frac{\mathrm{SE}(\hat{S}(24))}{\hat{S}(24)\log(\hat{S}(24))}\right]\right)$$

```
> cat("One-sided 95% lower confidence limit:", round(one_sided_LCL, 3), "\n")
One-sided 95% lower confidence limit: 0.603
```

Decision Rule

Reject H_0 if the lower bound of the one-sided 95% confidence interval for S(24) is greater than 0.40**.**

Fail to reject H_0 otherwise.

```
Reject H0: Survival at 24 months is significantly greater than 0.40
```

We repeated the process for the 5000 times and got the results as:

```
> # ----------------------------
> cat("Number of simulations        :", nsim, "\n")
Number of simulations        : 5000
> cat("H0 rejected (success)         :", n_reject, "\n")
H0 rejected (success)         : 4998
> cat("H0 accepted (failure)         :", n_accept, "\n")
H0 accepted (failure)         : 2
> cat("Observed probability of success:", round(prob_success, 3),
"\n")
Observed probability of success: 1
>
```

Conclusion

A simulation study consisting of 5,000 replicated trials was conducted to evaluate whether the study drug improves 24-month survival compared to the standard of care. Under the assumed true 24-month survival probability of 60%, the null hypothesis H0: S(24)=0.40 was rejected in 4,998 out of 5,000 simulated trials using a one-sided 95% confidence interval based on the Kaplan–Meier estimator.

The observed probability of success was therefore approximately 1, indicating that the proposed study design has very high power to detect a clinically meaningful improvement in 24-month survival. These results suggest that, under the stated assumptions, the study drug is highly likely to demonstrate statistically significant superiority over the standard of care.

Layman conclusion

We simulated the clinical trial 5,000 times assuming the drug truly improves survival. In almost every simulation, the drug showed a clear survival benefit compared to current treatment. This means the planned study is very likely to succeed in showing that the new drug is better, provided the assumptions hold true.

APPENDIX:

```
# ----------------------------------------------------------------------
# Hackathon NMIMS'26 - Survival Analysis Project
# ----------------------------------------------------------------------

# ========================
# Exercise 1: Data Preparation
# ========================

library(readr)      # CSV reading
library(lubridate)  # Date interval calculations

# Load original dataset
Dataset <- read_csv(" /Dataset.csv")

# Standardize date columns
Dataset$recruitment_date <- as.Date(Dataset$recruitment_date)
Dataset$event_or_withdrawal_date <- as.Date(Dataset$event_or_withdrawal_date)

# Calculate follow-up duration in months
# NA if patient still ongoing (no event/withdrawal date)
Dataset$diff_months <- time_length(
  interval(Dataset$recruitment_date, Dataset$event_or_withdrawal_date),
  "month"
)

# Binary: 1 = follow-up < 24 months (early failure possible)
#         0 = ≥ 24 months or still ongoing
```

```
Dataset$more_than_24_months <- ifelse(
  !is.na(Dataset$diff_months) & Dataset$diff_months < 24,
  1, 0
)

# Binary: 1 = event of primary interest (PD or Death)
#         0 = withdrawal, missing, other
Dataset$reason_cat <- ifelse(
  Dataset$reason %in% c("Disease Progression", "Death"),
  1, 0
)

# Very strict composite event:
# 1 = early (<24m) AND disease progression/death
# 0 = everything else (including all censored + late events)
Dataset$final <- ifelse(
  Dataset$more_than_24_months == 1 & Dataset$reason_cat == 1,
  1, 0
)

# Quick verification
colnames(Dataset)

# Save processed version
write_csv(Dataset, "/New_Final_Dataset.csv")


# ========================
# Exercise 2: Single Simulation Example
# ========================

set.seed(125)        # Fixed for reproducibility in report

N <- 200             # Sample size
analysis_time <- 48  # Study end time

# 1. Uniform accrual 0–24 months
entry_time <- runif(N, 0, 24)

# 2. Exponential event times targeting 60% EFS at 24 months
lambda_event <- -log(0.60) / 24
event_time <- rexp(N, lambda_event)

# 3. Low dropout rate (≈5% per year)
lambda_drop <- 0.05 / 12
dropout_time <- rexp(N, lambda_drop)

# 4. Administrative censoring at study end
admin_censor_time <- analysis_time - entry_time

# 5. Observed time = first of competing events
```

```
observed_time <- pmin(event_time, dropout_time, admin_censor_time)

# 6. Event indicator (1 = failure observed, 0 = censored)
event_indicator <- as.numeric(event_time <= pmin(dropout_time, admin_censor_time))

# Create simulation dataset
sim_data <- data.frame(
entry_time       = entry_time,
event_time       = event_time,
dropout_time     = dropout_time,
  admin_censor_time = admin_censor_time,
  observed_time    = observed_time,
  event            = event_indicator
)

# Basic diagnostics
head(sim_data)
table(sim_data$event)
summary(sim_data$observed_time)

# Kaplan-Meier estimation
library(survival)
library(survminer)

km_fit <- survfit(Surv(observed_time, event) ~ 1, data = sim_data)

# Plot with risk table and 24-month reference line
km_plot <- ggsurvplot(
  km_fit,
  conf.int = TRUE,
  censor = TRUE,
  risk.table = TRUE,
  xlab = "Time since entry (months)",
  ylab = "Event-free survival probability",
  title = "Kaplan-Meier - Simulated Single-arm Trial (n=200)",
  ggtheme = theme_minimal()
)

km_plot$plot <- km_plot$plot +
  geom_vline(xintercept = 24, linetype = "dashed", color = "red")

print(km_plot)

# 24-month point estimate and confidence interval
s24 <- summary(km_fit, times = 24)
cat("24-month EFS estimate:", round(s24$surv, 3), "\n")
cat("95% CI:", round(s24$lower, 3), "-", round(s24$upper, 3), "\n")
```